

Problem

Can we use gravity as an efficient source of renewable energy?

On a large scale, gravitational energy storage is projected to be 80-90 percent efficient. For perspective, solar and wind power are 20-40 percent efficient, the two most common renewable energy sources being used. The goal of this experiment is to determine whether or not gravitational power can be efficiently generated on a small scale. This innovation can be implemented to common applications such as a street lamp.

Hypothesis

If the mass being dropped increases, then the production of electrical energy due to gravity will increase

Potential energy varies directly with mass, meaning that as mass increases, potential energy increases, allowing for more energy to be converted into electricity. Furthermore, the kinetic energy of a moving object increases as the mass of the object increases. The drop of the weight will have more kinetic energy when it has more mass; therefore, spinning the motor faster and generating more electricity.

Materials

Multimeter

4000 RPM DC Motor

30.48 Meters of Nylon Rope

10 kg Barbell Plates

5 kg Barbell Weights

Bulb Socket

Block and Tackle Pulley

Spool With Inner Opening

Ball Bearings

Stackable Gearbox

Red and Black Wires

60 Watt Light Bulb

Breadboard

Soldering Iron

Metal Rod

Procedure

1. Gather the materials, put on the safety goggles, and insulated gloves
2. Hang the pulley at a height of 3 meters from ground level using a secure structure or frame.
3. Insert metal rod into spool's inner diameter using a clamp to hold spool in place in order to hammer safely
4. Thread rope through the pulley and wind the excess to the spool
5. Get 3 meters of rope, insert it through the holes of the appropriate barbell plates for each trial and tie the ends together
6. Hook the weight onto the pulley using the rope that is tied to the weight on ground level
7. Assemble the gearbox and then attach the shaft of the gearbox to the rod of the spool using the coupling
8. Assemble the breadboard circuit using the wires, motor, and battery
9. Using the battery, power the motor to pull up the weight 3 meters high (This is a stand-in for a solar panel which would power the motor)
10. Use the multimeter to measure the voltage and current that is needed to lift the weight
11. After the weight reaches the top, detach the battery from the breadboard, add the light bulb to the circuit, and rewire the circuit so that the motor will generate electricity for the bulb
12. The weight will then be lowered slowly which will rotate the spool, then the gears, and finally the motor to illuminate the light bulb
13. When the weight is dropped, use the multimeter to track the voltage and current while the weight is dropping
14. Repeat steps 9-13 for each trial of different weights (IV) while ensuring that the barbell is pulled up to three meters in each trial.

Background Research

- **Record player powered by dropping a water jug over a cliff**
- **Light bulb powered for 2 hours after lifting for 30 seconds**